

RAJIT GOEL

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Education

Georgia Institute of Technology

Master of Science in Computer Science | Artificial Intelligence Specialization

August 2026 – December 2027

Atlanta, GA

- On-Campus Program

University of Texas at Dallas

Bachelor of Science in Computer Science | GPA: 3.965/4.0

August 2022 – May 2026

Richardson, TX

- CS Honors Program
- Certificate in Applied Experience Design and Research
- Goldman Sachs Excellence in Computer Science Scholarship recipient
- Capital One Scholarship in Applied Experience Design and Research recipient

Professional Experience

Samsung Electronics America

June 2026 – August 2026

Software Engineer Intern, Networks, Systems Support / Tools & Automation

Plano, TX

- Owned the Optimus AI Assistant end-to-end, used by 40+ field engineers: an on-premise, retrieval-augmented agentic assistant that lets RF and field engineers query equipment manuals, live network configuration, and operational databases in natural language, covering agent orchestration, retrieval, tool integration, the web interface, and Docker/nginx deployment
- Built the orchestration layer in Python/Django with LangGraph: a 15-node state machine (intent detection → context retrieval → query resolution → routing) branching into knowledge-base RAG, MCP tool invocation, or database-query paths, plus a ReAct orchestrator bounded by two-tier iteration and tool-call budgets
- Replaced a binary context check with a ternary relevance grader (sufficient / refine / absent) that triggers another retrieval round or returns no answer when the retrieved context is insufficient
- Integrated hybrid semantic + full-text retrieval over a LanceDB vector store (682 documents, BGE-M3 embeddings) and built a figure-retrieval pipeline that surfaces diagrams from source PDFs alongside text answers, using vision-model descriptions generated at ingest time
- Built the natural-language-to-SQL path with read-only enforcement, parsing generated T-SQL with sqlglot and validating the root expression against an allowlist of read-only statement types
- Created and maintained viaero-mcp, a Model Context Protocol server exposing 100+ read-only diagnostic tools and NETCONF/XML configuration retrieval as shared infrastructure for two independent production AI systems; permission gating is driven by server-declared ToolAnnotations
- Designed the conversation-memory and context-budgeting layer, and root-caused a defect where the compaction limit resolved at import time, compacting long conversations against an 8,192-token budget instead of the model's actual 131,072
- Built the React / TypeScript / Redux chat interface, including message branching (a conversation tree where editing or regenerating forks a navigable branch) and a live agent-transparency UI surfacing which tool ran, what was retrieved, and what SQL executed
- Diagnosed and fixed production streaming defects (nginx proxy read timeout, SSE heartbeat for slow first-token generation), and root-caused a third to Django's StreamingHttpResponse buffering under a WSGI gthread worker, handed off as an executable runbook
- Built an LLM gateway with health-checked failover across three on-prem NVIDIA GPU hosts, adding per-host model inventory so requests route only to hosts that hold the requested model
- Co-designed the architecture of the AI Alarm System, a LangGraph plan-execute-replan agent for automated alarm triage that cuts field-engineer diagnosis time from ~2 hours to 10–20 minutes, and built the MCP tool layer its execute stage runs on
- Drove release engineering for the subsystem, consolidating five long-lived feature branches into a single reviewable release and verifying zero regressions against a measured control run
- Built and optimized Apache Airflow DAGs powering automated ticket-analytics dashboards, restructuring task dependencies to parallelize independent stages and eliminate redundant work between tasks
- Developed daily in Python on remote Linux servers over SSH under air-gapped constraints (no runtime internet, CDNs, or hosted APIs), using Docker, Docker Compose, nginx, and gunicorn/daphne for multi-service deployment
- Built a Python evaluation harness comparing 9–10 self-hosted models on answer quality, tool-call validity, and latency against a 4–5 minute SLA to select the production model; authored a 20-slide deck and Confluence handover

Cinemark Theatres

June 2025 – August 2025

Data Science Intern

Plano, TX

- Analyzed theater demographic data (income, ethnicity, age) using Python (pandas, scikit-learn) and SQL to identify trends in Food & Beverage performance across 300+ locations
- Built K-Means clustering models to segment theaters and compared key KPIs (per cap, incidence rate, occupancy) across clusters to inform pricing and promotional strategies
- Conducted targeted release analysis to assess the impact of specific movie launches on F&B spending behavior across demographic segments

- Performed market basket analysis with SQL and Excel to uncover high-value product pairings and upsell opportunities, improving combo design recommendations
- Designed and delivered executive presentations with data visualizations in Power BI, Excel, and matplotlib, translating complex findings into actionable business insights
- Collaborated cross-functionally on "Pop & Top" initiative, leveraging guest surveys and international sales data to develop a business case for topped popcorn offerings

Thomson Reuters

May 2024 – August 2024

Software Engineer Intern

Frisco, TX

- Developed RESTful APIs in C# ASP.NET for Form 1040 validation, building a base later reused for other tax forms, and cutting processing time by 10–15%
- Refactored validation logic into modular C# components, reducing turnaround from weeks to near real-time
- Queried large datasets in SQL Server Management Studio, ensuring accurate integration of complex validation rules
- Converted legacy XML rules into modern XSLT, streamlining integration with current platforms
- Collaborated with a global Agile/Scrum team (Canada, India, Mexico, Russia), managing Git-based version control for reliability

Projects & Research

Abnormal Latte

July 2026 – Present

Lead Programmer, *Unity 6 Third-Person Action RPG (title under NDA)*

Remote

- Architected the narrative systems (branching dialogue, quests, character relationships, and time-of-day progression) and authored the technical design documentation for a 20+ person cross-discipline team
- Ran a build-vs-buy evaluation of narrative middleware, selecting and integrating an open-source dialogue runtime over a bespoke implementation and cutting an estimated 4–6 weeks of engineering scope
- Designed an in-editor authoring tool letting non-technical writers build branching dialogue in a node graph, with plain-text files as the source of truth and compiler-backed validation to catch content errors at import
- Reviewed cross-discipline design documentation, surfacing specification conflicts and unowned system dependencies ahead of implementation
- Built and deployed the studio's automation service in Python, running 24/7 as a systemd service on an Oracle VM, with 20 commands spanning meeting recording, discussion boards, server backups, a world clock, and per-person timezones
- Engineered its meeting recorder to write one Opus track per speaker, removing the need for diarisation, splitting tracks over the transcription upload cap at the longest silence, and isolating per-track failures so one bad track does not lose the others
- Built a two-pass LLM pipeline that turns transcripts into structured meeting minutes delivered to Google Drive, plus a decision-record system for forum discussions mirrored to a Google Sheet
- Worked around platform limits by updating a pinned message for the world clock, since Discord caps channel renames at 2 per 10 minutes, and stored IANA zone names so daylight saving stays correct

Appeara - AI-Powered VR Survival Game | *Unity 6, C#, OpenXR, OpenAI API, Sentry*

September 2026

- Built a VR survival game at Hack the North 2026 where players describe or sketch a weapon and fight with it while defending a besieged space station, placing Top 10 in the Sentry track
- Designed the weapon pipeline: OpenAI turns spoken descriptions, sketches, and combat context into a structured WeaponSpec under a strict JSON schema, which Unity validates against a gameplay balance budget before assembling projectile, beam, thrown, melee, or bow weapons
- Built an adaptive enemy system where the alien Mothership analyzes each fabricated weapon and selects countermeasures, starting that analysis at fabrication so model latency no longer overran the 20-second adaptation window
- Generated weapon-specific firing sounds and character dialogue with ElevenLabs, cached to avoid repeat requests, with local rule-based fallbacks that keep the game playable when an AI provider is unavailable
- Traced the full interaction with Sentry across microphone capture, transcription, weapon generation, Unity assembly, and audio, using performance traces to find frame-time spikes that led to object pooling; telemetry excluded raw audio, transcripts, and credentials

SONARE.LIVE - Real-Time Lyric Performance Game | *Three.js, MediaPipe, WebGL/GLSL*

April 2026 – June 2026

- Designed and built a browser-based, fully client-side real-time rhythm game for the Hatsune Miku "Magical Mirai 2026" Programming Contest, spanning ~85 ES modules with no UI framework or backend by design
- Engineered a dual-input gesture engine that unifies mouse drags and live webcam fingertip tracking into a single input stream, feeding a custom shape-recognition pipeline (normalized template matching, arc-length resampling, and scored matching) that auto-tunes difficulty per input mode
- Built a real-time hand-tracking pipeline using MediaPipe (Pose + Hand Landmarkers) with 1-Euro filtering, median despiking, and asymmetric hysteresis for reliable pen-up/pen-down detection and two-hand tracking without label-swapping
- Developed a 4-layer procedural character animation system driving VRM blendshapes (viseme lip-sync, idle blink/gaze, music-driven emotion, and pointer-relative gaze) with Three.js, custom GLSL water shaders, and spring-bone physics
- Moved MediaPipe inference off the main thread into a Web Worker with createImageBitmap frame transfer and a capability-gated fallback, protecting 3D render FPS on low-end hardware while degrading gracefully to a full mouse-only experience
- Authored a portable static build pipeline (PowerShell) that rewrites absolute asset paths to relative via regex and emits forward-slash zip entries, enabling deployment from any sub-path host (e.g., itch.io) and correct extraction on Linux servers

Blockchain Messenger - E2E Encrypted Messaging | *React, Node.js, Express, Web Crypto API*

May 2026

- Built a fully end-to-end encrypted, peer-to-peer messaging application on a custom proof-of-work blockchain implemented entirely from scratch, using no third-party blockchain or cryptography libraries
- Implemented client-side hybrid encryption (AES-256-GCM wrapped with RSA-OAEP) and ECDSA P-256 message signing via the browser Web Crypto API, ensuring private keys never leave the client
- Developed a Node.js + Express node tier with a REST API for transaction submission, SHA-256 proof-of-work mining, chain validation, and longest-valid-chain consensus across a decentralized peer network
- Enforced deterministic transaction and block validation rules (signature verification, identity binding, replay protection, PoW linkage) as the system's "smart contract" layer, backed by a 27-test automated suite

ArrestorIQ - Flame Arrestor Analysis Platform | *React Native, Expo, Electron, TypeScript, Haxe* **January 2026 – May 2026**

- Built a cross-platform desktop and mobile application for Emerson (industry-sponsored capstone) that unifies flame arrestor test, CFD simulation, and bill-of-materials data for 10,000+ product configurations under a single validated platform
- Engineered a search and filtering system integrating uFuzzy fuzzy-matching across all product attributes, running $\sim 5\times$ faster than Emerson's internal proprietary search while supporting discrete and numeric-range filters
- Architected a unified filter pipeline that applies discrete and range filters before fuzzy matching on the narrowed candidate set, keeping the UI responsive with the full catalog loaded
- Designed and implemented the multi-product comparison feature and sortable, filterable table views used across the catalog and evaluation screens for side-by-side regression and pressure-drop analysis
- Led design of the team's project poster and architecture diagram, translating technical contributions for a non-technical audience of faculty and industry sponsors

Eukarya - Creature Evolution Game | *Unity, C#, Blender, Figma* **January 2026 – May 2026**

- Developed a creature evolution simulator spanning six evolutionary stages with both 2D top-down and full 3D environments, including a transition mechanic that lets the player breach the water's surface onto land
- Programmed the camera system, UI/UX, and audio architecture, including a persistent DontDestroyOnLoad music manager with string-keyed track lookup that loops themes seamlessly across scene transitions
- Implemented health, stamina, and overhealth mechanics using an edge-detection state machine to fire attack damage exactly once per instance independent of script execution order
- Built a counter-based PlayerPrefs unlock and level-select system and resolved cross-platform build issues (editor-vs-build hitbox bugs, HUD fog-tinting via a dedicated UI camera, aspect-ratio letterboxing)

PreFace - Redesigning the Hiring & Onboarding Experience | *Figma, React, UX Research* **August 2025 – December 2025**

- Partnered with Fisher Investments for ATCM 4341 (Applied Experience Design capstone) to reimagine recruitment by replacing traditional cover letters with interactive, role-specific courses
- Designed dual-sided platform: applicants complete skill-building modules earning verifiable certificates; employers review detailed assessment breakdowns across technical proficiency, soft skills, and values alignment
- Conducted user research, developed personas, created high-fidelity Figma prototypes, and built React frontend components for applicant dashboard, course progression, certificate sharing, and HR portal
- Enabled actionable hiring insights through structured assessments and match scoring, reducing candidate apathy and bias in traditional resume-based screening

Project Pawkour | *Unity, C#, UI Design, Figma, OpenGL* **August 2025 – December 2025**

- Built a third-person parkour game where players control a cat escaping from a secret laboratory through a low-poly environment
- Developed custom C# scripts for physics-based movement mechanics including running, jumping, dashing, and wall-running
- Implemented camera system with smooth following and adaptive positioning based on player movement and velocity
- Designed complete UI system including menus, HUD elements, and interactive components; scripted all UI functionality and interactions
- Created storyboards defining game narrative, level progression, and player experience flow from start to escape
- Integrated adaptive music system that intensifies based on player velocity, enhancing immersion during high-speed parkour sequences

Aegis - Privacy Shopping Browser Extension | *JavaScript, Chrome API, Supabase* **August 2025 – December 2025**

- Built a web extension for CS 4352 (HCI) that aggregates e-commerce reviews across the web, computes trust & review scores, and uses those scores to conditionally autofill or withhold sensitive user information
- Implemented popup UI, content scripts, and background service worker (Manifest v3) to inject safety indicators into shopping flows and surface visual trust cues (e.g., charts for risk levels)
- Integrated RESTful API with Supabase (PostgreSQL, authentication, RLS) for storing and retrieving review/usage data
- Designed HCI-focused UX flows including clear risk feedback, consent affordances, and data usage history views

AI-Powered Whiteboard - Ideate | *React, Next.js 16, TypeScript, PostgreSQL, NVIDIA Nemotron API* **November 2025**

- Engineered an AI-powered ideation platform that transforms hand-drawn sketches and written concepts into structured product blueprints, including flowcharts, business pitches, competitive analyses, and 90-day roadmaps
- Integrated the NVIDIA Nemotron API (vision + text models) using a dual-model architecture to interpret sketches, extract features and user flows, and generate coherent product narratives
- Built an interactive whiteboard in React, Canvas API, and perfect-freehand, enabling precise drawing, shape recognition, undo/redo, and real-time manipulation of visual elements
- Developed Next.js 16 + TypeScript REST APIs for AI workflows including sketch analysis, Mermaid diagram generation with syntax validation and retry logic, automatic pitch creation, and competitor market analysis

- Designed PostgreSQL schemas with Supabase for persistent canvases, Row Level Security (RLS), and Google OAuth-based session management
- Deployed the full-stack application on Vercel using serverless functions for AI endpoints, optimizing performance and minimizing latency for sketch-to-insight generation
- Award: Earned top-5 recognition (Honorable Mention, NVIDIA Track) out of 100+ projects at HackUTD 2025 for innovative multimodal AI integration and end-to-end product ideation automation

AI Sound Effects Generator - Auralis | *Next.js, Flask, PyTorch, PostgreSQL, Gemini API* **June 2025 – November 2025**

- Created an AI sound generation platform that produces studio-quality effects from text prompts, powered by PyTorch AudioLDM latent diffusion models for real-time synthesis with CUDA acceleration and CPU fallback
- Integrated Google Gemini 2.5 Flash API for adaptive prompt refinement, enhancing the diversity, realism, and coherence of generated audio outputs
- Built a Next.js 15 frontend with Supabase authentication, Row Level Security (RLS), and Google OAuth, providing personalized user sessions and secure content management
- Architected a Flask REST API backend optimized for low-latency streaming, blob storage, and CORS-compliant delivery, ensuring smooth cross-browser performance
- Designed PostgreSQL schemas with triggers, analytics tracking, and indexed queries supporting a public sound library, like system, and real-time engagement metrics

Designing for Real Inclusion at UTD - HCI Research | *Qualitative Research, Figma, UX* **January 2025 – May 2025**

- Investigated nonbinary students' campus experiences after 28% reported not feeling a sense of belonging
- Conducted qualitative interviews revealing students discovered resources only through informal networks and existing policies lacked enforcement
- Proposed interventions including a centralized LGBTQ+ resource hub/map, anonymous feedback system for misgendering incidents, identity change guide for campus systems, and inclusive event feed with notifications
- Presented findings and design recommendations to UTD faculty and administration, emphasizing shift from performative to structural support

Leadership and Involvement

Chinese Student Association (CSA @ UTD)

January 2025 – May 2026

Design Lead & Family Head

Richardson, TX

- Led creative direction for one of UTD's largest cultural organizations, designing in Figma t-shirts purchased by 200+ members and creating social media posts featured across CSA's Instagram
- Mentored "fam" group of first-year students, coordinating bonding activities, competitions, and collaborative initiatives across 50+ members

HackMate

January 2025 – May 2025

Frontend Engineer

Richardson, TX

- A web platform designed to connect hackathon participants, enabling team formation, project collaboration, and event networking
- Developed frontend components using React, CSS, and Figma to deliver a consistent and engaging UI/UX
- Implemented navigation menus, group management tools, and contact forms integrated with backend APIs for dynamic content

Nebula Labs

August 2023 – January 2024

UI Designer

Richardson, TX

- Designed and iterated UI prototypes in Figma for Jupiter, a platform connecting students with campus organizations
- Partnered with developers and peers to translate user research into intuitive flows, ensuring designs aligned with technical feasibility and product goals

STEMpump

June 2020 – September 2022

Head of Media

Remote

- Led branding and marketing for a student-run STEM education nonprofit
- Created and taught a Python AI & TensorFlow course; produced educational videos and social media content
- Managed a media team and coordinated with developers to maintain consistent branding across all platforms

Technical Skills

Languages: C#, C/C++, Python, Java, JavaScript/TypeScript, R, SQL, Bash, HTML/CSS, MIPS Assembly, Verilog, XML, XSLT

Frameworks & Libraries: Django, Django REST Framework, ASP.NET, React.js, Redux Toolkit, Next.js, Node.js, Flask, PyTorch, TensorFlow, scikit-learn, pandas, matplotlib, Three.js, Unity

AI / ML: Retrieval-Augmented Generation (RAG), LangGraph state machines, ReAct agentic loops, LLM agents, Model Context Protocol (MCP), Ollama, self-hosted LLM inference, vector databases (LanceDB), embeddings (BGE-M3), vision-language models, structured output / JSON-Schema-constrained generation, context budgeting, LLM evaluation harnesses, prompt engineering, MediaPipe

Developer Tools: Git, GitHub Enterprise, Docker, Docker Compose, nginx, gunicorn/daphne, Apache Airflow, Linux (SSH / remote administration), Jira, Confluence, Azure, Azure DevOps, Vercel, Vite, SQL Server Management Studio, Sentry

Databases: PostgreSQL, SQL Server (T-SQL), MySQL, Redis, Supabase, LanceDB

Design Tools: Figma, Adobe Photoshop, Adobe Premiere Pro, AutoCAD

Other Skills: Test-driven development, pytest, Vitest, systematic root-cause debugging, release engineering, code review, technical documentation (ASD-STE100), Server-Sent Events, RESTful APIs, Agile/Scrum, Data Analysis, UI/UX Design, Interaction Design, Computer Networking, Operating Systems, Power BI, Excel, Claude Code, Cline, GitHub Copilot

Relevant Coursework

- Computer Science II
- Discrete Math I/II
- Linear Algebra
- Data Structures & Algorithms
- Computer Architecture

- Software Engineering
- Database Systems
- Operating Systems
- Digital Logic
- Artificial Intelligence

- Automata Theory
- Computer Graphics
- Human-Computer Interaction
- Interaction Design